

Student Name: Aditi

Project Title: Bookstagram

Progress Report Number: 1

Reporting Period: January 26, 2026 – February 8, 2026

Work Logs

Date	Number of Hours	Work Done
26-1-2026	1.5	Conducted analysis of backend requirements and designed the overall server-side architecture for the Bookstagram application
27-1-2026	2	Implemented the initial Node.js and Express.js setup and configured the server.js file for backend operations
29-1-2026	1.5	Established database connectivity and configured environment variables required for secure backend communication
30-1-2026	2	Developed the backend folder structure and created essential files including controllers, routes, models, and middleware components
1-2-2026	1.5	Implemented authentication controller logic to support user registration and login functionality
2-2-2026	1	Developed and integrated authentication routes with the corresponding controller logic
4-2-2026	1.5	Implemented middleware for request validation and route protection to enhance backend security
5-2-2026	1	Integrated frontend and backend communication flow for authentication APIs and

		tested request–response handling
7-2-2026	1	Designed and implemented backend workflow for AI-assisted book creation functionality, including required routes and file structures

Summary Description of Work Completed

During this reporting period, I focused on the backend development of the Bookstagram application. The starting work involved setting up the server, configuring database connectivity, and implementing user authentication functionality. Backend components such as controllers, routes, and middleware were developed to support secure user registration and login. Additionally, the integration flow between the frontend and backend was established to ensure proper API communication. During the development process, minor challenges were encountered, including initial difficulties in aligning request and response formats between the frontend and backend. These issues were resolved through incremental testing, code refactoring, and validation of API endpoints. No changes were made to the original project proposal. The next phase of development will focus on dashboard implementation and AI-assisted book creation features.

Repository Check-in of Implementation Completed

The following files and folders were checked into the GitHub repository during this reporting period:

- Implementation/backend/server.js
- Implementation/backend/config/db.js
- Implementation/backend/controllers/authController.js
- Implementation/backend/routes/authRoutes.js
- Implementation/backend/middlewares/authMiddleware.js
- Implementation/backend/models/User.js

Code was committed periodically.

AI Use Declaration

AI Tool name	Version / Account type	Description
ChatGPT (OpenAI)	GPT-5, Free Version	backend architecture and overall project workflow for Bookstagram
ChatGPT (OpenAI)	GPT-5, Free Version	project scope and defining functional requirements
ChatGPT (OpenAI)	GPT-5, Free Version	backend setup, including server configuration and database connectivity
ChatGPT (OpenAI)	GPT-5, Free Version	Assistance in understanding authentication flow using controllers, routes, and middleware
ChatGPT (OpenAI)	GPT-5, Free Version	Assistance in resolving implementation errors

Value Added to the Project Beyond AI Usage

- Designed and implemented the complete backend setup independently, including server configuration, database connection, and file structure.
- Built and tested user authentication features such as signup and login through hands-on coding and debugging.
- Manually connected frontend and backend components and resolved issues through testing.

Appendix

“How should I structure my backend folders in Node.js and Express?”

“How do controllers, routes and middleware work together in authentication?”

“How do I properly connect frontend API calls with backend routes?”

“How do I make a simple signup API in Node and Express where I can check if user already exists, validate name, email and password, create the user, and return a JWT token after successful registration?”

“How do I generate JWT token after user signup in Express?”